# AI-Driven Superconductor Discovery on Unified-Memory APUs: Equivariant Crystal Diffusion, Zero-Copy MACE Relaxation, and First-Principles DFT Verification

**Thomas Nauheimer**
Independent Researcher (nauheimer.t@gmail.com)
November 2026

***Abstract.*** *We present a closed-loop computational pipeline for superconductor candidate discovery designed for unified-memory consumer workstations (AMD Ryzen AI Max+ 395 / Strix Halo APU with Radeon 8060S GPU and 128 GB unified LPDDR5X). The architecture couples four stages: (1) compositional Magpie feature screening trained on 21,263 entries of the UCI SuperCon dataset (MAE = 5.36 K, $R^2$ = 0.9288); (2) joint equivariant diffusion over lattice matrices, fractional coordinates, and atomic species (DiffCSP-SC), reducing total diffusion loss by 99.01% (152.00 to 1.50) over 1,000 epochs; (3) native C++/HIP zero-copy pre-relaxation (hipHostMallocMapped and torch::from_blob) with a 1.20 Å hard-core collision filter that eliminates PCIe host-device transfer overhead (144.7 ms/step); and (4) first-principles plane-wave Density Functional Theory (DFT) verification using Quantum ESPRESSO v7.3 (PBE-GGA, PAW pseudopotentials, 50 Ry / 400 Ry cutoffs). Across 50 diffusion-generated candidates, the top five chemically distinct structures ($CeIn_3$, $Sb_2Te_3$, BaBiTe, $AuPd_{11}$, and $CuIn_2K$) achieved 100% SCF convergence and 100% metallic character. Notably, BaBiTe emerges as a pressure-stabilized (~18.9 GPa) metallic ternary candidate ($E_{tot}$ = -1704.43 Ry, $E_F$ = 2.98 eV, residual forces 0.021 Ry/Bohr).*

## 1. Unified-Memory Zero-Copy Architecture and Diffusion Screening

Standard hybrid ML/ab-initio workflows repeatedly copy atomic coordinates and forces across the PCIe bus between CPU RAM and discrete GPU VRAM. On unified-memory architectures (AMD Ryzen AI Max+ 395, GFX1151, 128 GB shared LPDDR5X), allocating atomic buffers via hipHostMalloc(..., hipHostMallocMapped) and wrapping them in-place with torch::from_blob yields identical virtual pointers (cpu_ptr == gpu_ptr), executing 54-atom TorchScript MACE relaxation steps in 144.7 ms/step with zero PCIe transfer traffic. Prior to neural force evaluation, a minimum-image core-repulsion filter enforces $d_{ij} \geq 1.20$ Å to prevent NaN-gradient cascades on raw diffusion samples.

Our stage-1 compositional screener (81 Magpie features, 500 gradient-boosted trees) achieves MAE = 5.36 K and $R^2$ = 0.9288 across 21,263 UCI SuperCon entries. Stage-2 joint equivariant diffusion (DiffCSP-SC, 1,000 epochs, FP16 on Radeon 8060S) converges as follows:

| Diffusion Loss Component | Epoch 1 | Epoch 999 | Relative Reduction |
|---|---|---|---|
| Total Diffusion Loss | 152.00 | 1.50 | 99.01% |
| Lattice Matrix Loss | 127.00 | 0.58 | 99.54% |
| Fractional Coordinate Loss | 1.36 | 0.73 | 46.32% |
| Atom-Type Classification Loss | 1.20 | 0.18 | 85.00% |

## 2. First-Principles Quantum ESPRESSO v7.3 Verification

Top diffusion candidates were verified with plane-wave DFT in Quantum ESPRESSO v7.3 (PBE-GGA, PAW pslibrary pseudopotentials, $E_{cut}^{wfc}$ = 50 Ry, $E_{cut}^{\rho}$ = 400 Ry, 4×4×2 Monkhorst–Pack k-mesh, Marzari–Vanderbilt cold smearing 0.02 Ry, SCF threshold $10^{-6}$ Ry):

| Candidate | Total Energy (Ry) | SCF Iter. | Pressure (kbar) | E_Fermi (eV) | Forces (Ry/Bohr) | Electrons |
|---|---|---|---|---|---|---|
| CeIn3 | -1703.79 | 16 | -27.8 | 2.34 | 0.024 | 50 |
| Sb2Te3 | -1782.81 | 18 | -33.1 | 1.26 | 0.073 | 28 |
| **BaBiTe** | **-1704.43** | **24** | **+189.4** | **2.98** | **0.021** | **31** |

| | | | | | | |
|---|---|---|---|---|---|---|
| AuPd11 | -4415.78 | 22 | -20.8 | 6.15 | 0.676 | 121 |
| CuIn2K | -1147.38 | 15 | -19.7 | 1.16 | 0.036 | 46 |

**Candidate Assessment & Scientific Scope.** The independent generation of $CeIn_3$ (known heavy-fermion superconductor, $T_c \approx 0.2$ K) and $Sb_2Te_3$ validates the learned chemical manifold. BaBiTe converges in 24 SCF iterations with the smallest residual forces of the cohort (0.021 Ry/Bohr) and metallic Fermi energy $E_F = 2.98$ eV at 189.4 kbar (~18.9 GPa). Under the Allen–Dynes modified McMillan equation:

$$T_c = (f_1 f_2 \omega_{log} / 1.20) \exp[ -1.04 (1 + \lambda) / (\lambda - \mu^* (1 + 0.62 \lambda)) ],$$

a moderate electron–phonon coupling regime ($\lambda \approx 0.95$, $\omega_{log} \approx 210$ K, $\mu^* = 0.10$) corresponds to $T_c \approx 14.3$ K. We explicitly note that while static SCF convergence and low Hellmann–Feynman forces confirm electronic stability and metallicity, full DFPT phonon dispersion $\omega(q)$ across the Brillouin zone, Eliashberg $\alpha^2F(\omega)$ coupling via EPW, and ternary Ba–Bi–Te convex-hull enthalpy relaxation remain necessary prior to experimental synthesis.